

SARRA BEN DOUB

Data Scientist & AI Engineer · Financial Data · Risk · AI

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PROFILE

Engineering graduate from ENSIT in applied mathematics and modelling, with experience in Data Science and AI Engineering across financial and insurance environments. I build Machine Learning, Generative AI and intelligent document processing solutions for financial data, risk and analysis problems. I am looking for an opportunity in Data Science, AI Engineering or Machine Learning in banking, insurance, leasing or fintech in Tunisia.

PROFESSIONAL EXPERIENCE

BIAT — Financial Document Intelligence & Credit Risk Feb – Jul 2026

Banque Internationale Arabe de Tunisie · AI Engineer & Data Scientist — Final-year project

- **Document validation & extraction.** Checking the nature of the document, then extracting financial tables from the statements filed as PDFs with the CMF using Docling and OCR (balance sheet, income statement, cash-flow statement), structured and stored in MongoDB.
- **Data cleaning & verification.** Cleaning, normalising and checking the extracted data (OCR errors, inconsistent values), then assigning it to the chart of accounts.
- **Financial & credit analysis.** Streamlit decision-support application computing liquidity, solvency, profitability and leverage ratios, with year-on-year comparison.
- **LLM-generated summary** with levels of assessment (favourable, reserved, unfavourable, needs further review), the final decision remaining with the analyst. Pipeline orchestrated with n8n, containerised with Docker.

Tech: Python · Docling / OCR · MongoDB · Streamlit · LLM · n8n · Docker

GAT Assurances — Claims Intelligence & NLP Jun – Aug 2025

Insurance · AI & Full-Stack Development · Internship

- **Claims-management platform:** public website, customer area and back-office, designed with a graphic designer for the visual identity and user experience.
- **Intent classification** through a French-language rule engine (missing accents, typos), backed by zero-shot classification with BART-large-MNLI when no rule decides.
- **Conversational assistant** collecting the missing information and creating the claim (category, priority, tracking number) routed to the back-office.

Tech: Python · Flask · Transformers · BART-large-MNLI · NLP · SQLite · HTML/CSS/JS | github.com/SarraBendoub/insurance-agency-platform

VALUE — Data Engineering & Analytics Jul 2024

Data Science · Data Engineering · Internship

- **Multi-source collection** automated with Scrapy and Selenium, then cleaning, structuring and merging the data into a usable dataset.
- **K-Means segmentation** and interactive Folium mapping to visualise how sites are distributed and spot areas with low digital visibility.

Tech: Python · Scrapy · Selenium · Pandas · scikit-learn · K-Means · Folium | github.com/SarraBendoub/HeritageScope

BNA Bank — Customer Service Chatbot Apr – May 2023

Banking · NLP & Machine Learning · Internship

- **Intent classification** on French-language queries: linguistic preprocessing, training and comparison of two architectures, FNN and LSTM.
- **Automated answers** from the bank's FAQ base (banking services, BNA eBanking, Mandat Cash), deployed with Flask.

Tech: TensorFlow · Keras · NLTK · Python · Flask

PERSONAL PROJECTS — FINANCIAL AI & RISK

CreditGuard — Explainable & Monitored Credit Scoring 2026

Credit risk · Probability of default

- **LightGBM** probability-of-default model, **AUC 0.757**, with **SHAP** explainability per prediction.
- **Drift monitoring** with **PSI**, Kolmogorov-Smirnov and Chi-squared tests to detect distribution shifts. Streamlit dashboard, Docker packaging.

Tech: Python · LightGBM · SHAP · SciPy · Streamlit · Docker | github.com/SarraBendoub/CreditGuard

FraudGuard — Fraud Detection & AML/CFT Monitoring 2026

Fraud · Financial Crime · Transaction Monitoring

- **XGBoost** fraud detection on **6.4M transactions** at **0.13% fraud**: **PR-AUC 0.89**, **F1 0.82**, threshold tuned on the trade-off between missed frauds and false alerts.
- **Transaction-monitoring rule engine** (velocity, structuring, account patterns), **BLOCK / REVIEW / ALLOW** action levels, SHAP explainability on alerts.
- **Data leakage detected and fixed**: an early version reached **PR-AUC 0.9998** using features posterior to the transaction, since removed.

Tech: Python · XGBoost · SHAP · pandas · Streamlit | github.com/SarraBendoub/FraudGuard

TECHNICAL SKILLS

Financial Data & Risk	Credit Risk · Probability of Default · Financial Statements · Fraud Detection · Model Monitoring · Explainable AI
Machine Learning	Python · Scikit-learn · XGBoost · LightGBM · SHAP · PyTorch · TensorFlow · Statistics
Data	SQL · PostgreSQL · MongoDB · Pandas · NumPy · Web scraping (Scrapy, Selenium, BeautifulSoup)
Document AI	OCR · Docling · Document Processing · LLM
Generative AI	RAG · Transformers · LLM · Ollama · Groq
Deployment	Docker · FastAPI · Streamlit · n8n · Git
Reporting & Data Viz	Power BI · Matplotlib · Seaborn · Streamlit dashboards

EDUCATION

Engineering degree — Applied Mathematics & Modelling, ENSIT	2023 – 2026
BSc — Data Science, FST	2020 – 2023
Baccalaureate — Experimental Sciences	2020

CERTIFICATION & LANGUAGES

Microsoft Azure AI Fundamentals (AI-900)
Arabic native · French fluent · English intermediate (TOEIC) · Italian A1